# REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

**Generated:** 2026-08-25 17:18:52

## 1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

| Metric | Value |
|---|---|
| Total Network Flows | 2744 |
| Normal Flows | 2579 |
| Detected Attacks | 165 |
| Attack Rate | 6.01% |

## 2. Machine Learning Model

| Parameter | Value |
|---|---|
| Algorithm | Random Forest Classifier |
| Training Features | duration, protocol_type, src_bytes, dst_bytes, count |
| Classification | Normal / Attack |
| Optimized Threshold | 0.30 |
| Validation Accuracy | 80.96% |
| Attack Precision | 0.93 |
| Attack Recall | 0.72 |
| Attack F1-Score | 0.81 |

## 3. Network Protocol Distribution

| Protocol | Flow Count |
|---|---|
| TCP | 1668 |
| UDP | 1074 |
| OTHER | 2 |

# 4. Detected Attack History

| Time | Source | Destination | Protocol | Confidence |
|---|---|---|---|---|
| 2026-08-25 17:18:23 | 10.2.0.1:65432 | 192.168.1.18:59821 | TCP | 42.68% |
| 2026-08-25 17:02:46 | 10.2.0.1:65432 | 192.168.1.18:57957 | TCP | 42.68% |
| 2026-08-25 17:01:40 | 150.171.110.85:443 | 192.168.1.18:65035 | TCP | 48.45% |
| 2026-08-25 17:01:39 | 10.2.0.1:65432 | 192.168.1.18:62778 | TCP | 42.68% |
| 2026-08-25 17:01:17 | 192.168.1.18:64452 | 8.8.4.4:443 | UDP | 48.98% |
| 2026-08-25 17:01:16 | 192.168.1.18:58934 | 202.88.211.77:443 | UDP | 31.79% |
| 2026-08-25 17:00:31 | 10.2.0.1:65432 | 192.168.1.18:51706 | TCP | 50.96% |
| 2026-08-25 17:00:20 | 192.168.1.18:51564 | 4.150.223.108:443 | TCP | 42.2% |
| 2026-08-25 16:28:06 | 10.2.0.1:65432 | 192.168.1.18:55386 | TCP | 42.68% |
| 2026-08-25 16:28:04 | 172.217.17.206:443 | 192.168.1.18:64258 | UDP | 85.5% |
| 2026-08-25 16:28:04 | 192.168.1.18:5353 | 224.0.0.251:5353 | UDP | 36.71% |
| 2026-08-25 16:28:02 | 10.2.0.1:65432 | 192.168.1.18:49874 | TCP | 42.68% |
| 2026-08-25 16:27:47 | 192.168.1.18:58127 | 3.111.241.224:443 | TCP | 92.55% |
| 2026-08-25 16:27:32 | 192.168.1.18:58127 | 3.111.241.224:443 | TCP | 35.49% |
| 2026-08-25 16:27:30 | 192.168.1.18:50919 | 4.224.112.242:443 | TCP | 41.2% |
| 2026-08-25 16:27:22 | 192.168.1.18:60443 | 23.205.101.152:443 | TCP | 35.83% |
| 2026-08-25 16:26:20 | 192.168.1.18:49672 | 23.211.60.116:443 | TCP | 36.6% |
| 2026-08-25 16:25:46 | 192.168.1.18:63265 | 8.8.8.8:443 | TCP | 88.91% |
| 2026-08-25 16:25:43 | 192.168.1.18:57464 | 57.144.211.32:443 | TCP | 93.41% |
| 2026-08-25 16:25:39 | 192.168.1.18:58786 | 8.8.4.4:443 | TCP | 88.91% |

# 5. Security Recommendations

**1.** Investigate repeated attack predictions from the same source address.

**2.** Monitor network flows with unusually high traffic volume or packet counts.

**3.** Review suspicious TCP and UDP connections identified by the ML classifier.

**4.** Use the attack confidence score together with network context when validating alerts.

**5.** Continue collecting representative network traffic to improve the IDS model.

**6.** Periodically retrain and evaluate the model using updated datasets.

# 6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.